

Machines augmenting entrepreneurs: Opportunities (and threats) at the Nexus of artificial intelligence and entrepreneurship[☆]

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ABSTRACT

Artificial intelligence (AI) refers to machines that are trained to perform tasks associated with human intelligence, interpret external data, learn from that external data, and use that learning to flexibly adapt to tasks to achieve specific outcomes. This paper briefly explains AI and looks into the future to highlight some of AI's broader and longer-term societal implications. We propose that AI can be combined with entrepreneurship to represent a super tool. Scholars can research the nexus of AI and entrepreneurship to explore the possibilities of this potential AI-entrepreneurship super tool and hopefully direct its use to productive processes and outcomes. We focus on specific entrepreneurship topics that benefit from AI's augmentation potential and acknowledge implications for entrepreneurship's dark side. We hope this paper stimulates future research at the AI-entrepreneurship nexus.

Executive summary: Artificial intelligence (AI) refers to machines that are trained to perform tasks associated with human intelligence, interpret external data, learn from that external data, and use that learning to flexibly adapt to tasks to achieve specific outcomes. Machine learning is the most common form of AI and largely relies on supervised learning—when the machine (i.e., AI) is trained with labels applied by humans. Deep learning and adversarial learning involve training on unlabeled data, or when the machine (via its algorithms) clusters data to reveal underlying patterns.

AI is simply a tool. Entrepreneurship is also simply a tool. How they are combined and used will determine their impact on humanity. While researchers have independently developed a greater understanding of entrepreneurship and AI, these two streams of research have primarily run in parallel. To indicate the scope of current and future AI, we provide examples of AI (at different levels of development) for four sectors—customer service, financial, healthcare, and tertiary education. Indeed, experts from industry research and consulting firms suggest many AI-related business opportunities for entrepreneurs to pursue.

Further, we elaborate on several of these opportunities, including opportunities to (1) capitalize on the “feeling economy,” (2) redistribute occupational skills in the economy, (3) develop and use new governance mechanisms, (4) keep humans in the loop (i.e., humans as part of the decision making process), (5) expand the role of humans in developing AI systems, and (6) expand the purposes of AI as a tool. After discussing the range of business opportunities that experts suggest will prevail in the economy with AI, we discuss how entrepreneurs can use AI as a tool to help them increase their chances of entrepreneurial success. We focus on four up-and-coming areas for entrepreneurship research: a more interaction-based perspective of (potential) entrepreneurial

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opportunities, a more activities-based micro-foundation approach to entrepreneurial action, a more cognitively hot perspective of entrepreneurial decision making and action, and a more compassionate and prosocial role of entrepreneurial action. As we discuss each topic, we also suggest opportunities to design an AI system (i.e., entrepreneurs as potential AI designers) to help entrepreneurs (i.e., entrepreneurs as AI users).

AI is an exciting development in the technology world. How it transforms markets and societies depends in large part on entrepreneurs. Entrepreneurs can use AI to augment their decisions and actions in pursuing potential opportunities for productive gains. Thus, we discuss entrepreneurs' most critical tasks in developing and managing AI and explore some of the dark-side aspects of AI. Scholars also have a role to play in how entrepreneurs use AI, but this role requires the hard work of theory building, theory elaboration, theory testing, and empirical theorizing. We offer some AI topics that we hope future entrepreneurship research will explore. We hope this paper encourages scholars to consider research at the nexus of AI and entrepreneurship.

1. Introduction

Stephen Hawking (2016) wrote of artificial intelligence (AI) that it “will be either the best or the worst thing ever to happen to humanity. We do not yet know which”. While the jury is still out, AI is simply a tool, and how people decide to use it will determine its impact on humanity. Indeed, the same could be said for entrepreneurship—it is a tool used for productive, unproductive, and destructive outcomes (Baumol, 1996). Combining AI with the entrepreneurial process could provide a super tool with the potential for even greater variety in outcomes. A tool can lead to destructive consequences when people do not know how to use the tool, how the tool works, or what the full implications of using the tool will be. For example, Amazon used a recruiting algorithm that inadvertently led to discriminatory recruiting practices. Therefore, the widespread understanding of these tools—AI and entrepreneurship—can help nudge their use toward more productive outcomes and away from destructive outcomes and prepare entrepreneurs for their role in monitoring and quickly resolving unintended consequences.

While researchers have developed a greater understanding of entrepreneurship and AI independently, these two streams of research have primarily run in parallel (for exceptions see Chalmers et al., 2021; Obschonka and Audretsch, 2020; Townsend and Hunt, 2019). However, as alluded to above, combining these tools is potent in generating impactful outcomes. This paper explores their intersection by focusing on how AI (the machine) can augment entrepreneurs (the humans) in pursuing potential opportunities for productive gains for themselves and society and in minimizing the potential dark side of destructive entrepreneurial action.

2. How AI works

AI refers to machines that are trained to perform tasks associated with human intelligence—that is, interpret external data, learn from that external data, and use that learning to flexibly adapt to tasks to achieve specific outcomes (adapted from Haenlein and Kaplan, 2019; He et al., 2019). As this definition implies, AI involves learning, and learning is based on algorithms. Algorithms are simply methods for computing things (Berk, 2021). As described below, these algorithms can be developed by people and machines.

Fig. 1 illustrates how machine learning AI works to develop and modify algorithms (adapted from Kane et al., 2021). The basic model of machine learning AI involves four major components: (1) data inputs ([a] predictor inputs and [b] outcomes inputs) lead to a (2) model (parameters, weights, and inter-relationships), which generates an (3) outcome (that a human designer wants to identify), which in turn generates (4) feedback to the data inputs.

AI technologies are designed to identify specific outcomes chosen by human designers, such as the accuracy of categorizing dogs and cats based on features displayed in a photo. This outcome needs to be concrete and measurable but can involve multiple sub-outcomes. To begin the machine learning process for identifying specific outcomes, a human designer needs to decide what data is available that reflects the range of possible inputs that a machine learning system could pattern match and includes the outcomes of interest. This data is referred to as training data. For example, for accurately categorizing dogs and cats, the training data would need to include photos of the animals, tags specified by a human indicating which images include a dog or a cat, and then a range of features in the images (e.g., different ear angularities, different head angularities, and so on) that help distinguish cats from dogs. Based on this training data, the system identifies even more features that may improve the accuracy of distinguishing between cats and dogs and includes model weights to increase categorization accuracy. These model weights may involve many different parameters and complex inter-relationships that determine accuracy (e.g., a five-way interaction). As AI operates, it generates new experiences (inputs and outcomes) that provide feedback for the AI system to adapt to the desired outcomes (or further enhance). In other words, AI learns and adapts to that learning. There are different learning modes used in AI, the most common of which is machine learning.

2.1. Machine learning for AI

Machine learning generates knowledge for computers—namely, based on patterns among data, the machine learns from experience (i.e., from data that is input and collected over time via feedback). This machine learning AI requires a specified search space, which typically includes objectives for the patterns sought, such as accuracy in categorizing dogs and cats. It also depends on the accuracy and size of the training data since improper training data (e.g., LinkedIn data about prior Amazon recruits) may be biased, thereby perpetuating

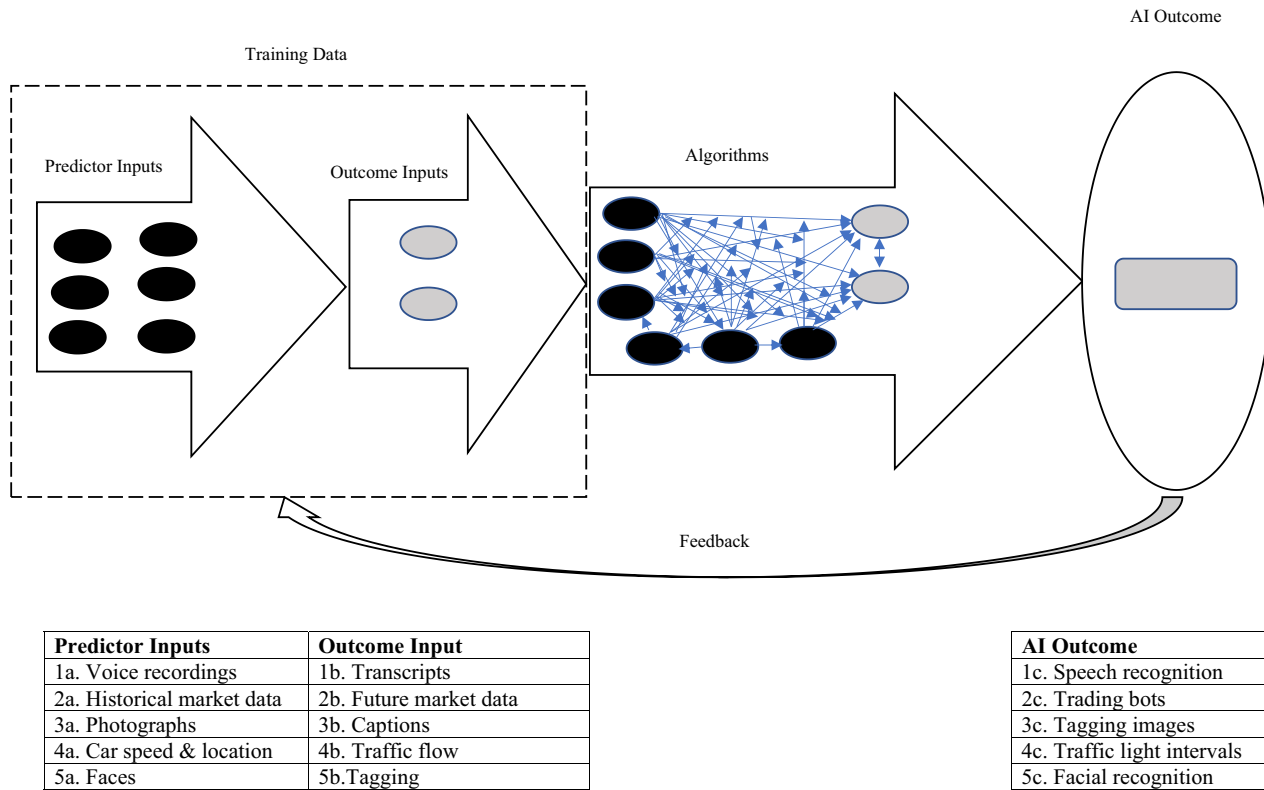


Fig. 1. An entrepreneur-based illustration of how machine learning AI works.

future patterns. Thus, at this point, machine learning is most useful with highly replicable patterns. For example, AI has been utilized with inputs of voice recordings and outputs of transcripts for speech recognition, inputs of historical market data and outputs of future market data for trading bots, inputs of photographs and outputs of captions for tagging images, inputs of car speed and location and outputs of traffic flow for setting traffic lights, and inputs of faces and outputs of tagging for facial recognition (Brynjolfsson and McAfee, 2017; Jordan and Mitchell, 2015). AI updates rapidly as new data focused on optimizing the specific outcomes becomes available via experience, but changes are typically incremental as adaptations are made (Amershi et al., 2014). For example, Netflix's movie recommendations for a person may change after the person watches a non-recommended movie but only slightly.

Other examples are those from the Healthcare industry in which machine learning AI is rapidly progressing. We offer two illustrations of how AI is used to create value for its users. First, AI is being used to quickly and accurately distinguish lesions from nodules in people's lungs for more effective treatment. For instance, the Arteys AI system finds lesions based on lung scans (CT scans) and liver scans (CT and MRI scans). This lesion-detection system serves a valuable role in augmenting physicians' work to save people's lives and offers other improvements in health outcomes (i.e., not treating people for possible lesions when they do not exist) (He et al., 2019). As a second illustration, AI is being used to provide advance notification of the probability of blood glucose changes so that individuals can prepare an insulin injection or take other preventative measures to avoid the negative effects of such changes, such as "vomiting, excessive hunger and thirst, rapid heartbeat, [and] vision problems," among others (Cleveland Clinic, 2022). The Guardian Connect AI system, for instance, warns individuals via their phones of major fluctuations in their blood glucose levels 10–60 min in advance based on changes observed in other parts of their bodies (He et al., 2019).

Although *machine learning* is the most common form of AI, improvements in quantum computing are leading to *deep learning* and *adversarial learning* becoming more common (Berente et al., 2021).¹ Indeed, the above examples of machine learning AI mostly rely on supervised learning—that is, when the machine is trained with labels applied by humans. In contrast, deep learning and adversarial learning involve training on unlabeled data, or when the machine (via its algorithms) clusters data to reveal underlying patterns (He et al., 2019).

2.2. Deep learning AI

Deep learning AI does not require a human to label the training data but uses algorithms to extract features from raw data without human inputs. Bini (2018: 2360) provides the following examples to illustrate how deep learning works in contrast to the machine learning described above:

Google's algorithms can use deep learning-based facial recognition to identify the same person (Uncle Bob) in 2 photographs taken a decade apart without any human input. Although it was never told that Uncle Bob is the one with the blond hair and crooked nose, if you identify that person as "Uncle Bob," the software can go and find him in all your other pictures (or any on the web). But AI can go many steps further. Google DeepMind's Deep Q-learning software can learn how to play Atari's Breakout video game without being told anything other than to maximize the score (Mnih et al., 2013). It is not told what a ball or a bat is. It has no concept of tennis or ball games. And yet, after a series of trial-and-error games from which it learns that to maximize the score, it must hit the "ball" with a "bat" to knock out some targets, in a matter of minutes, it can play at the level of a very skilled person and achieve very high scores. And if that were not enough, it can take those learned skills and apply them to other video games. Shocking to everyone except the folks at Google, in May of 2016, Google's program called AlphaGo beat the Korean GO master at the most complex game known to man (Silver et al., 2016). However, as with humans, the software is only as good as the data it is trained on. A person trained as a lawyer is not likely to be a great physician or a good plumber. Similarly, the Atari playing software DeepMind would not be all that great at face recognition, and I doubt AlphaGo is all that great at selecting the right medications for patients.²

2.3. Adversarial learning AI

Adversarial learning is an autonomous AI technology (i.e., it can learn without initial human inputs like deep learning) based on multiple deep neural networks. A neural network is a "structured framework with multiple processing layers that attempt to model high-level abstractions of the data" (Liu and Laili, 2018: 183). An adversarial learning AI typically involves two neural networks—one as a generator and the other as a discriminator (Goodfellow et al., 2014). The generator neural network constructs some conceptualization from the data (consistent with a tentative model generated through abduction) that is then tested using the discriminator neural network to identify and evaluate patterns in large datasets (Berente et al., 2021). While adversarial networks are primarily used to defend systems from attacks, they are starting to be used for other purposes, including medical imaging (Özdenizci et al., 2020), coherent human-like dialogue responses (Ren et al., 2020), and video game design (Seidel et al., 2020) (see Berente et al., 2021).

¹ Quantum computing, "unlike a classical computer which operates on 1 s or 0 s (bits), can operate in a superposition of these states called qubits ... [exceeding] every capability of the classical computers currently available" (King, 2018: 5).

² In deep learning, AI creates the labels (rather than humans as in machine learning) and then works to predict the labels to refine the deep network (Bengio et al., 2021).

3. AI as a phenomenon of interest to entrepreneurs

AI is widely used for various purposes, and these purposes will expand as AI advances. For example, Meitu (a start-up) offers a mobile phone app that makes people look more beautiful in selfies; Dataminr (another start-up) helps companies make better marketing and stock market decisions based on information gathered from monitoring the Internet and social media; and the infamous Cambridge Analytica used Facebook data to assess users' personalities to better tailor political messages to them (Kaplan and Haenlein, 2019).

We offer Table 1 to demonstrate the scope of current and future AI. Table 1 (adapted from Kaplan and Haenlein, 2019, and other sources) provides examples of AI (at different levels of development) for four sectors—customer service, financial, healthcare, and tertiary education.

In the customer service sector, AI has been used to personalize messages to customers, automatically respond to customers' inquiries (using chatbots), respond to customers' adverse emotional reactions to improve the customer experience (via facial recognition), customize service (e.g., using robots that speak most languages), and eventually increase delivery speed (by anticipating purchases) and security (e.g., through voiceprint). For example, some online stores use AI to introduce an avatar—an icon or figure representing a person or character—to enhance the customer experience. For instance, when a customer arrives on a firm's webpage, an avatar can offer a voice-enabled human-like interface that attracts the customer's attention, encourages the customer to ask questions, provides the customer directions around the online store, highlights product promotions, and provides recommendations (Brown, 2021). Such avatars have been shown to increase customer engagement and improve firms' productivity (Brown, 2021). Entrepreneurs can identify and construct opportunities to enhance customer service through AI-based products. They can also use these products to learn more about customer needs (as a source of new opportunities); improve venture operations in terms of speed, customization, quality, and reliability; and reduce payment risks with more secure payment methods.

In the finance sector, AI is used to detect and respond to financial fraud, identify investment opportunities in financial markets, and customize risk assessments. For example, CoinGenius is an “AI-driven cryptocurrency trading platform that gives crypto traders AI-based scoring systems, high-quality data feeds, and advanced forecasting to personalize news and trading features to maximize their efforts” (Daley, 2019: np). Entrepreneurs can develop new products using AI for the financial sector, such as to provide more accessible and more effective authentication for financial instruments, more immediate customized responses to financial clients, improved security using the blockchain, and more help in making decisions under uncertainty. Entrepreneurs can also use these products (current and forthcoming) in their ventures to protect against fraud, access debt capital for venture growth, provide customers increased financial security, and improve their ventures' efficiency and quality.

In healthcare, AI is used for diagnosis, treatment recommendations, treatment-outcome predictions, and robotics (e.g., robots performing surgery) that improve firm performance and patient outcomes. Entrepreneurs can identify and exploit opportunities to use AI in these products and services in the healthcare sector. Although most people think of large public or government hospitals in the healthcare sector, entrepreneurs also own and manage healthcare businesses. These entrepreneurial businesses can enhance their patients' experiences with AI. Entrepreneurs in the healthcare sector can improve decision-making reliability (e.g., consistent diagnosis and prognosis), speed (using real-time patient information), and customization (using personalized information) by implementing AI. Indeed, compassion venturing in healthcare is enhanced by AI through the improved scale and scope of customized speedy responses to alleviate people's suffering. One example of AI in the healthcare industry is BurstIQ, which created a ‘Health Wallet,’ combining AI, blockchain, and big data to holistically manage a patient's data. The Burst IQ wallet gives a patient's team of healthcare professionals access to his or her health records and wellness plans. These healthcare professionals can then choose to buy, sell, or trade patient data for different scientific studies or to learn more about specific diseases. However, the blockchain lets patients keep their identifying information private while sharing only big-picture health data (Daley, 2019: np).

In tertiary education, AI enhances students' learning experiences through virtual teaching assistants, personalized learning programs and interventions, and improved teacher performance. Teacher performance is enhanced by using AI for analytical tasks, such as checking for plagiarism, grading exams, and ensuring online students pay attention, thereby allowing teachers to focus their resources on mentoring, coaching, and moderating discussions. For example, one professor used AI to answer students' questions:

At the Georgia Institute of Technology, many of the students in a master's-level AI class were unaware that one of their teaching assistants, Jill Watson, wasn't human. (This was despite the clue in her name, which refers to IBM's Watson.) The class of approximately 300 students posted about 10,000 messages a semester to an online message board, a volume nearly impossible for a regular assistant to handle. . . . [The] class's professor, Ashok Goel, [noted] that while “the number of questions increases if you have more students . . . the number of different questions doesn't really go up.” So, he and his team created a system that could respond to those queries that were consistently repeated, and released Jill onto the message board. They populated Jill's memory with tens of thousands of questions (and their answers) from past semesters. Not only did most students not realize Jill was virtual, she was also among the most effective teaching assistants the class had seen, answering questions with a 97% success rate, according to Slate. Jill had “learned to parse the context of queries and reply to them accurately” (Neelakantan, 2020: np).

Again, entrepreneurs can develop AI products for the education industry. Education can also help entrepreneurs become more effective at identifying and exploiting opportunities, starting up a new organization, growing their ventures, and creating societal wealth. Entrepreneurship professors have the chance to adopt new AI-based educational services to provide a superior learning experience to naïve, nascent, and current entrepreneurs.

Table 1

Examples of AI development across sectors.

Ai	Sector examples	Entrepreneurial context
Customer Service Sector		
Basic	-Screening resumes in human resources to select candidates for positions and tasks -Developing more personalized messages about products, services, and firms for marketing -Automatically responding to customer inquiries via chatbots	-Conducting product development -Recruiting venture members -Growing ventures -Identifying opportunities from inquiries
Advanced	-Verbally responding to customers via phone calls with chatbots -Identifying the emotions of customers in line for intervention (e.g., opening a new register) using facial recognition -Responding to customers with robot servers that speak 150 languages -Providing insights into effective sales calls to enhance training and customer service via speech analytics -Suggesting the “best” answers to agents via bots listening to service calls	-Conducting product development -Improving ventures' customer service -Learning from customers to improve operations and trigger new opportunities -Collecting information to improve employees' reliability and performance
Most advanced	-Shipping items to customers before they have placed an order -Using avatars for the online shopping experience -Authenticating customers with voiceprint and biometry	-Conducting product development -Enhancing ventures' speed -Ensuring the security of payments to ventures
Financial sector		
Basic	-Protecting from, detecting, responding to, and predicting financial fraud -Detecting market trends and identifying investment opportunities -Calculating comprehensive credit scores for borrowers -Determining banks' risk of delinquencies and default by cardholders	-Conducting product development -Protecting ventures from fraud -Enhancing access to debt to fund growth
Advanced	-Providing financial investment advice via robo-advisors -Identifying erratic financial transactions as early signs of dementia via robo-advisors -Providing unsupervised credit-card fraud detection -Enabling financial authentication through recognition of faces, voiceprint, and optical characters	-Conducting product development -Monitoring venture members' work transactions -Improving security for ventures and customers
Most advanced	-Offering meaningful and coherent dialogue responses for financial advice via chatbots -Providing proof of stake for the blockchain -Offering security for a “nothing at the stake” blockchain attack -Enabling prediction under uncertainty	-Conducting product development -Improving efficiency and quality -Expanding opportunities using blockchain -Aiding decisions under uncertainty
Healthcare sector		
Basic	-Identifying lesions from scans of livers and lungs -Predicting treatment outcomes -Providing diagnostic recommendations consistent with clinic guidelines and the most up-to-date reviews of medical research	-Conducting product development -Enhancing performance for health ventures -Improving the reliability of ventures' decision making
Advanced	-Linking a patient's medical record with “biomedical research databases, genome sequencing databanks, pathology laboratories, insurance claims, and pharmacovigilance surveillance systems, [and] data collected from mobile Internet of Things (IoT) devices such as heart rate monitors [to] generate clinically relevant information in real time for health professionals” (Lysaght et al., 2019: 300)	-Conducting product development -Adapting decision making in real time -Enhancing compassion venturing by improving speed and customization to alleviate patient suffering -Improving venture performance.

(continued on next page)

Table 1 (continued)

Ai	Sector examples	Entrepreneurial context
Most advanced	-Detecting, showing, and responding to emotions via robots as companions for the elderly (and the lonely) -Applying precision diagnostics to degenerative diseases -Enabling robotic surgery for non-routine operations	-Conducting product development -Enhancing ventures' efficiency and reliability -Enabling social ventures to help more
Tertiary education sector		
Basic	-Providing AI-based virtual teaching assistants -Facilitating teaching and learning via chatbots -Checking for plagiarism or misuse of statistics	-Conducting product development -Educating entrepreneurs -Researching entrepreneurship
Advanced	-Analyzing facial expressions to determine if students are paying attention during virtual classes -Automatically launching interventions based on students' personal attributes and progress -Helping teachers develop more effective teaching strategies -Catching students cheating on exams -Grading essay-style exams	-Conducting product development -Educating entrepreneurs -Freeing up time from the analytical tasks of teaching (e.g., grading and monitoring) to enable more value-add activities (e.g., mentoring students and researching entrepreneurship)
Most advanced	-Providing learning interventions that identify students' emotions and respond with emotional content -Coaching, moderating, and facilitating discussion -Addressing issues of equity, ethics, privacy, and data ownership	-Conducting product development -Educating students about entrepreneurship -Training entrepreneurs' emotional intelligence

4. Entrepreneurial opportunities with AI

Experts from industry research and consulting firms, such as Gartner and Forrester, suggest many business opportunities for entrepreneurs to pursue based on the expected integration of AI across industries. Below, we elaborate on several opportunities for entrepreneurs to leverage AI or address the needs of others who implement AI.

4.1. The opportunity to capitalize on the feeling economy

The previous economy can be considered the mechanical economy, and the current economy can be considered the thinking economy. The mechanical economy is primarily based on “mechanical/physical/repetitive tasks,” such as inspecting equipment, scheduling work, moving objects, controlling machines, operating equipment, storing information, and maintaining electronic equipment (Huang et al., 2019: 45). AI is prevalent and effective in performing many mechanical tasks that are typically repetitive and require reliable performance (e.g., Huang and Rust, 2018). For example, AI has been implemented in factory automation to enhance production efficiency.

The current thinking economy involves dealing with information, including identifying processes and objects, processing and evaluating information, organizing work, monitoring and controlling processes, making decisions, and updating knowledge (Huang et al., 2019). In this economy, AI involves automatic learning and adaption from ongoing experience (data generation). For example, autonomous cars reflect thinking in their interpretations of and responses to driving conditions, drivers, and vehicles. Future AI in the thinking economy will include higher forms of intelligence, such as intuitive thinking (Schoenick et al., 2017).

As AI performs many of the tasks of the mechanical economy and more of the functions of the thinking economy, Huang et al. (2019: 44–45) propose that we will enter into a feeling economy emphasizing the “interpersonal, empathetic, feeling tasks of a job.” These tasks include communicating with members at different organizational levels; establishing interpersonal relationships within and across organizational boundaries; caring for others; issue selling to influence managers' decision making and organizational action; negotiating and resolving conflict; building teams; motivating others; and coaching, training, and developing personnel (Huang et al., 2019). Early AI advancements in the feeling economy include chatbots (e.g., Alexa and Siri) that engage others in response to a feeling rather than an informational request; technology that monitors a driver's emotional and attentive state for autonomous driving; and robots with more emotional, physical, and voice displays, including human-friendly robotics entertainment, companion robots, and service robots.

Future advancements in AI for the feeling economy will include recognizing, communicating, and responding to emotions. First, future AI will be more effective in recognizing human-generated emotions using massive textual/visual databases of emotions from physical sensors and personal devices that collect physiological data on one's own and others' emotions. This AI will also harness data from analyzing images, videos, and audio recordings to detect facial features and voice-tone indicators of emotions and will model the level of emotional activity in teams (see Franzoni et al., 2019). Second, future AI will incorporate more emotions to enhance communication by using physical sensors and personal devices to provide physiological data on one's and others' emotions (see Franzoni et al., 2019) and to learn from advances in emotional recognition to improve communication. Finally, AI advancements in recognizing and communicating emotions will enable robots to enhance humans' quality of life by forming stronger human-robot connections, displaying empathy when rehabilitating patients, and improving communication for training and education purposes (including education about emotions). Therefore, future AI will provide stronger emotional connections both between humans and machines and between humans themselves.

Although AI has substantial potential for the future, machines do not yet have human senses, emotions, and social skills (Braga and Logan, 2017; Kaplan and Haenlein, 2019; Raisch and Krakowski, 2021). Entrepreneurship often involves social tasks, such as co-constructing potential opportunities (Seyb et al., 2019), engaging resource holders (Fisher et al., 2020), and using social skills to make sales (Baron and Markman, 2000). Entrepreneurship often entails noticing, understanding, and even regulating one's own and others' emotions and behaviors (Cardon et al., 2012; Foo, 2011). For example, the entrepreneurial process is typically an emotional rollercoaster (Shepherd, 2003; Shepherd and Cardon, 2009), so entrepreneurs must understand and regulate their own and others' emotions to make good decisions and manage relationships with key stakeholders to ensure access to essential resources (Cardon et al., 2012). Entrepreneurship can thus provide the human touch, emotions, and high-quality relationships to augment AI for enhanced decision making and the effective implementation of AI.

4.2. The opportunity to redistribute occupational skills in the economy

AI transforms businesses by generating opportunities that create value for users. For example, AI reduces the diagnostic time for intracranial hemorrhage by 96%, thereby saving lives and reducing disabilities (McKinsey and Company, 2018). Indeed, experts expect the adoption of AI to lead to relatively little job abolition (5% according to McKinsey and Company, 2018; this is currently disputed by Zanzotto, 2019) because the activities recommended by the AI tool still need to be implemented by humans (e.g., to stop the bleeding or make policy decisions). However, humans' activities will be different, leading to some workers having to switch careers and others having to change how they perform tasks to work alongside and interact with AI systems. For example, the McKinsey Global Initiative report (2018) describes a consensus among experts that AI will impact 30% of the activities of 60% of careers such that by 2030, it is expected that 400–800 million workers will be performing substantially different jobs than they do now and 550–890 million new jobs will be created.

Occupations that will experience the greatest change in content and volume include managers, plumbers (in unpredictable physical

environments), teachers, high-skill medical professionals, and nursing aides. For entrepreneurs, then, facilitating these transitions will provide an enormous business opportunity.

4.3. The opportunity for new governance mechanisms

As AI transforms society, entrepreneurs will need to monitor and influence how AI is governed. Indeed, Zuboff (2019) describes how AI is a threat to our economy. It is a threat to our democracy because it has created monopoly companies that control what information we see and what products we buy. Accordingly, regulations are needed to address monopoly-related issues. In particular, regulations will be required to control how AI systems are trained and tested but will need to focus on the process of generating AI while allowing (unknown) future applications of AI to continue. For example, just as quality-related product and safety warranties are enforced through government regulations, Haenlein and Kaplan (2019) propose a similar process for AI algorithms to make organizations responsible for their AI systems.

But who regulates regulators? While China is working on AI to “allow the trustworthy to roam everywhere under heaven while making it hard for the discredited to take a single step” (The Economist, 2016: np), San Francisco has banned the use of facial recognition technology in the city (Conger et al., 2019). The European Union is taking steps to “get ahead” of some technology issues by introducing the General Data Protection Regulation, which stipulates how personal information can be collected, stored, and processed (Haenlein and Kaplan, 2019).

Another governance mechanism for overseeing and directing the development of AI is activism by the AI community. The AI community includes “researchers, research engineers, faculty, graduate students, NGO workers, campaigners, and some technology workers more generally.” This community engages in activism through a “broad range of social and political campaigning, organizing and advocacy ... [such as] issue-framing; agenda-setting; standard-setting; private discussions with decision-makers; public campaigning in traditional and social media; establishing new fields and organizations; submissions to governmental inquiries; and classic labour tactics of boycotts and strikes” (Belfield, 2020: 15). Kane et al. (2021) recommend that users become active in creating their own AI bots to increase their quality of life and protect them from the purely financial goals underlying the decisions to use AI.

This situation presents a business opportunity for entrepreneurs to develop and monitor third-party governance mechanisms.

4.4. The opportunity to keep humans in the loop

Experts argue that the future of AI and society is not one in which AI replaces the work humans do but augments it by providing complementary inputs into the tasks being performed by humans, referred to as “keeping humans in the loop.” For example, AI can augment the work of a journalist by providing a newsfeed to inform him or her of information for a story, with the newsfeed having learned from the journalist's prior selections. In this case, the journalist is kept in the loop by informing the AI based on his or her last story selections and then further informing the AI based on the adoption of newsfeed information (or not), like Netflix's machine learning from whether a person clicks on a recommended movie. An entrepreneur considering an AI system needs then to consider the needs of the humans using the AI system. For example, an entrepreneur might develop an AI system that augments a researcher's work in generating a literature search for a paper by learning which articles the researcher chose from Google Scholar and which sections he or she highlighted in those articles for further review and argument development. This assistance would be useful, of course, only if searching for articles was a non-random, completely replicable event. For some researchers, however, their creativity is enhanced explicitly by articles they do not expect to find, limiting the value of such an AI system (unless it includes some “outlier” articles to stimulate non-linear thinking). Ultimately, the opportunity exists for entrepreneurs to leverage AI in creating systems to support other individuals' ability to more efficiently and effectively fulfill key tasks.

4.5. The opportunity to expand the role of humans in developing the AI tool

The relationships humans have with AI supersede simply implementing AI recommendations. Humans decide on data collection and data management for the AI tool. This presents an opportunity for entrepreneurs to help expand the range of occupations that engage in such selection processes. Today, these humans may be low-level analytic staff who are not required to assess the bias and accuracy of datasets. Entrepreneurs can help ensure that those best suited to making these judgments are appropriately engaged. Low-level analytic staff are also involved in deciding on the outputs expected from the AI tool: are these outputs simply categorization accuracy, or are they more human oriented? However, limiting such essential decisions to humans who are not experts in the work process for which AI is being used may cause more damage than help. Therefore, entrepreneurs have an opportunity to expand the engagement of humans in deciding how outputs are displayed.

The machines will crunch any data for training purposes, yet the data may not be appropriate, accurate, ethically obtained, or contextually sensitive (Brynjolfsson and McAfee, 2014; Raisch and Krakowski, 2021). This data crunching requires entrepreneurs to become familiar with available datasets and the conditions under which each dataset was obtained, what data was not included, and whether the data opens these entrepreneurs (and others) to increased security and privacy concerns. For example, collecting current health information on individuals exposes entrepreneurs to the possibility of violating laws that protect the secrecy of patients' sensitive data if sold to other companies. While enforcement of data-selling standards is not yet sophisticated, entrepreneurs need to be aware of regulations and ethics related to the collection, use, and dispersion of data used in and generated by AI.

4.6. The opportunity to expand the purposes of the AI tool

Since the machines do not yet have a sense of purpose or intention, the first decision for entrepreneurs is to define the objective of AI in a manner that is both ethical and appropriate in relation to the limitations of AI (Raisch and Krakowski, 2021). Entrepreneurs can provide AI with objectives based on their personal purpose and intentions. Indeed, entrepreneurial action is typically purpose driven and intentional. For example, individuals enter entrepreneurship for various purposes, including to gain income (Åstebro and Chen, 2014; Carter, 2011), to increase their autonomy (Douglas and Shepherd, 2002; Shir et al., 2019), and to create something of beauty (Cova and Svanfeldt, 1993). Their entrepreneurial intentions are driven by these motivational drivers of the potential desirability of the outcomes of entrepreneurial action and by their assessments of the feasibility of successfully undertaking such action (i.e., given their entrepreneurial self-efficacy [Hsu et al., 2019, role models [Nowiński and Haddoud, 2019], and education [Liu et al., 2019]).

From a practical point of view, in the AI context, entrepreneurs will need to change the way they conceptualize “purpose” to include responsibility and accountability for producing AI and for its use and continued evolution (including future regulation). This responsibility and accountability are critically important yet challenging given the causal ambiguity of how AI generates outcomes, how AI systems change over time as new data is added (which makes any initial understanding quickly obsolete), and what unintended consequences may arise from using AI. Indeed, a significant critique of Mark Zuckerberg is his failure to accept any responsibility for the information silo wars on the Internet from using his social network (which is also a critique of the government for insufficient or outdated regulations).

The purpose of an AI tool must fit the domain in which it is applied. AI systems work for a specific task. Indeed, Krohn (2019) refers to current AI as artificial narrow intelligence because it can only perform narrow tasks. It is not yet possible for AI to display general intelligence to successfully perform across a range of functions (Berk, 2021). Entrepreneurs can augment AI by thinking across domains of knowledge. Specifically, entrepreneurs are often creative in generating novel solutions to problems (Ardichvili et al., 2003; Baron and Tang, 2011). These creative solutions could come from unobvious sources (Ward, 2004) or from innovatively combining and recombining resources that had previously been discarded or utilized for different purposes (Baker and Nelson, 2005) based on, for example, analogical reasoning (from other domains [e.g., Cornelissen and Clarke, 2014; Sergeeva et al., 2021) or structural alignment creativity (Grégoire et al., 2010; Grégoire and Shepherd, 2012). For example, computational creativity—a sub-field of AI—can develop a system that interacts with users (emotionally and cognitively) such that the machine and humans co-create something novel and useful, including art and design compositions (DiPaola, 2014) and music compositions (Feldman, 2017). Thus, practically, AI lends itself to augmenting humans' exposure to and collaboration across domains for creative outcomes. Research is needed on how the AI tool can be involved in what are fundamentally human activities as well as on what markets are appropriate for entrepreneurs with this tool.

5. AI augmenting entrepreneurs

In the previous section, we spoke about the range of business opportunities that experts suggest will prevail in the economy with AI. In this section, we flip the role of AI and discuss how entrepreneurs can use the AI tool to help them increase their chances of entrepreneurial success. In 2015, Shepherd was the editor-in-chief of the *Journal of Business Venturing* and proposed four up-and-coming areas for entrepreneurship research: (1) a more interaction-based perspective of (potential) entrepreneurial opportunities, (2) a more activities-based micro-foundation approach to entrepreneurial action, (3) a more cognitively hot perspective of entrepreneurial decision making and action, and (4) a more compassionate and prosocial role of entrepreneurial action. We focus on the first two topics because they are immediately applicable to understanding the intersection of AI and entrepreneurship—we discuss Topics 1 and 2 and suggest opportunities to design an AI system (i.e., entrepreneurs as potential AI designers) to help entrepreneurs (i.e., entrepreneurs as AI users). In the discussion, we speculate about the possibilities of Topics 3 and 4 informing our understanding of the AI-entrepreneurship relationship.

5.1. AI augmenting a more interactive perspective of entrepreneurship

Research on entrepreneurial cognition has traditionally focused on individual entrepreneurs and their perceptions of the external environment (Shepherd et al., 2017; Tornikoski and Newbert, 2007), prior knowledge of markets, ways to serve markets, and customer problems (Grégoire et al., 2010; Shane, 2000) and on the identification, evaluation, and decision to exploit potential opportunities (Choi and Shepherd, 2004; Haynie et al., 2009). Entrepreneurship scholars can understand how potential opportunities are identified and refined by taking a more interactive perspective. For example, Shepherd (2015) focuses on social interactions between a community of inquiry and an entrepreneur under the assumption that an idea for a potential opportunity likely comes from the entrepreneur's mind due to having experienced the world and then testing the potential opportunity back in the world. Testing the veracity of the potential opportunity can involve the community of inquiry—potential stakeholders—providing feedback about the plausibility of the potential opportunity (Autio et al., 2013; Seyb et al., 2019). This feedback informs and motivates the entrepreneur to modify the existing potential opportunity or abandon it. The community of inquiry provides feedback that changes the entrepreneur's mind, and the potential opportunity (and its various iterations) can change the composition and nature of the community of inquiry. Therefore, there is a mutual adjustment between the entrepreneur's mind and the community of inquiry, and the potential opportunity is the mechanism and product of that adjustment (which also adjusts). This co-construction of a potential opportunity is a critical but challenging task.

AI may be able to augment this interactive approach to entrepreneurship. Specifically, AI can augment entrepreneurs' ability to

notice potential opportunities. An entrepreneur's prior knowledge provides the basis for detecting something in the external environment that may represent a potential opportunity by limiting the aspects of the environment that he or she attends to in order to make such detection possible (given bounded attention and cognition). Entrepreneurship scholars have been interested in how this knowledge directs attention (Shepherd et al., 2017; Zahra and Chaples, 1993), provides a knowledge corridor (Shane, 2000; Venkatraman, 1997), creates filtered attention (Nicolini and Korica, 2021), and provides limited absorptive capacity for interpreting and understanding noticed changes (Domurath and Patzelt, 2016; Zahra and George, 2002). From a more macro-perspective, individuals have knowledge corridors to identify opportunities. The problem for entrepreneurs is recognizing, developing, and using their knowledge corridors to identify potential opportunities to exploit. One entrepreneurial venture, IntelligentX, uses AI to propose and refine a new line of beers. Namely, "The firm first deployed a bot to ask consumers about flavour preferences. It then fed the data into an algorithm that was used to create its first line of beer products. The recipes are continually tweaked based on ongoing feedback" (Morantz, 2021: np).

In a community of (independent or corporate) entrepreneurs (perhaps on a site devoted to entrepreneurial ventures in a vertical industry or projects within a large organization), opportunities identified by others that are ultimately not exploited might be stored as data that can later be matched to other entrepreneurs. Distinctions can be made between opportunity identifiers and opportunity exploiters like in collaborative crowdsourcing, in which the various participants play multiple different roles (Malhotra and Majchrzak, 2014). With accumulated data, an AI model can help the community recognize the need for more role-related activities, allowing individuals to self-select to pursue potential opportunities that the AI tool suggests might match their interests. For example, AI can offer specific partnerships of identifiers and exploiters for specific corridors (knowledge and motivation) of opportunity. This matching of different people and outcomes for various process stages then provides feedback to the AI system making the matches.

Furthermore, AI is useful for other forms of pattern recognition. AI has already made substantial progress in noticing and detecting patterns, including patterns in climate (Barnes et al., 2019), energy use in cities (Aviso et al., 2021), and images (e.g., convolutional neural networks [Stoel, 2019]). For example, AI is used for email security by learning from past emails to develop filters that sift through current emails to classify a threat (rather than an opportunity as in the discussion of entrepreneurs above), such as spam, phishing, and malware (or the level of security of each specific email) (Surya, 2017). AI has also been used to detect trends in disease (rather than in technology adoption for entrepreneurship). For example, HealthMap is a group (community of inquiry) of researchers, epidemiologists, and software developers who monitor informal sources of information about disease outbreaks to evaluate the threat to public health (Long and Ehrenfeld, 2020). Further, Helfrich (2022: np) provides an example of how entrepreneurs can more effectively perform market research to notice trends in market demand and more effectively promote their ventures' products or services:

In the fast-paced and ever-evolving digital-marketing landscape, getting a steady stream of new clients can be pretty challenging. The key to achieving maximum marketing success now depends on marketers' ability to deliver highly personalized, relevant buying experiences. . . . The adoption of artificial intelligence in digital marketing enables marketers to gain a more comprehensive understanding of their target audiences. . . . Machines can quickly identify patterns such as past buying history, buying preferences, credit scores, and other common threads. . . . [For example], using ML [machine learning]-trained recommendation models, market research can personalize market surveys more efficiently by presenting the relevant questions when the audience is most available and receptive. [AI] can analyze vast data way more efficiently than any human. And while doing so, it filters responses that don't suit the research requirements.

As mentioned, AI can match opportunity identifiers with opportunity exploiters. AI has already been used in matching, and advancements continue to be made. For example, AI is used to match workers with employers on a gig-economy platform (Liu et al., 2021), matching cancer patients (based on structured and unstructured text data in their electronic health records) to "complex eligibility criteria" for clinical trials (Jin et al., 2019; Woo, 2019), and matching the content of persuasive appeals to individual targets' psychological characteristics (Matz et al., 2017). These applications provide evidence that AI can be developed to match potential opportunities to people, match a single technology to different people to generate additional potential opportunities, and match identifiers with exploiters within a particular motivation and knowledge corridor. For example, Valuer.ai uses AI to match startups with established organizations. The startups benefit from the established organizations' funding, reputation, and networks, whereas the established organizations learn about the startups' innovations.

Valuer finds matches based on a number of factors from each party, including technology, team, management, vision, selling points, market, results, and much more. . . . The Copenhagen-based company is connected to almost 4000 technology scouts across 100 startup hubs around the world. The scouts regularly interview startups to extract information—and then, Valuer combines the findings from said interviews with findings from a random forest algorithm that predicts early stage startups' growth potential. As so, Valuer only includes the best possible startups on its platform. As of right now, 65 new startups are added every day. As for the corporations, they sign up to the platform through a subscription model in order to be matched with the startups. "When a corporation has been added to the platform, we use the information from our tech scouts as well as AI techniques to screen startups that live up to the corporation's request," Dennis Poulsen explains. "Then, the AI system automatically selects the startups that match best with the corporation and its needs." The result is a partnership with the best possible chance of being a success—for the corporation and for the startup alike. (Technology.org, 2019: np).

However, knowledge corridors may lead entrepreneurs to miss potential opportunities. Indeed, entrepreneurial ideas can begin in the minds of entrepreneurs based on their abduction from a noticed anomaly. Entrepreneurs find it difficult to determine whether an anomaly signals a potential opportunity or noise (given the overlap in the distributions of both [McMullen and Shepherd, 2006]). This

doubt about a signal of a potential opportunity represents entrepreneurs' uncertainty over whether to act on the noticed data or not. Of course, there are also likely signals of potential opportunities that remain undetected. AI can help entrepreneurs separate potential opportunity signals from noise and notice signals that would otherwise have remained hidden. This task is similar to when AI searches MRI and CT scans to distinguish lesions from nodules in lungs and livers (stimuli requiring action versus inaction) (He et al., 2019).

In detecting anomalies, Eisenhardt (2008); Eisenhardt and Tabrizi, 1995) highlights the importance of real-time information yet acknowledges that potential opportunities still depend on someone connecting the dots based on intuition. AI can be developed and used to scan real-time information to highlight anomalies and patterns of anomalies. This form of AI could involve supervised learning based on human designers' labels and would detect anomalies and patterns relative to the focal entrepreneurial expectations (i.e., expectations used to label the AI for supervised learning). Such an approach would be highly effective for incremental opportunities, consistent with a top-down approach based on an entrepreneur's prior knowledge, experiences, and expectations and input information from current users' feedback, signals of change in the current market, and movements by existing competitors. However, it would be less effective for noticing anomalies in aspects of the environment that are not expected to be important and are thus not labeled (Tripsas, 2009; Tripsas and Gavetti, 2000). These anomalies that entrepreneurs do not notice and are therefore not included in the AI could signal a discontinuous change in the external environment (Shepherd et al., 2017). Such signals of discontinuous change could include emerging technologies in previously unrelated industries (Eggers and Kaplan, 2009), architectural changes that alter how components of a product are integrated and linked together into a coherent whole (Henderson and Clark, 1990), and a new entrant at the low-cost end of a market (Kumar, 2006).

AI that uses unsupervised learning (e.g., deep learning AI) would better notice these more unanticipated anomalies. Deep learning AI provides the opportunity to induce patterns from unlabeled data that humans were not previously aware of (Jordan and Mitchell, 2015; Raisch and Krakowski, 2021). This noticing of unexpected patterns provides a more bottom-up approach for allocating an entrepreneur's attention by allowing the environment to speak to the entrepreneur (via AI) and trigger the entrepreneurial idea-generation process to offer the initial version of a potential opportunity. For instance, AI could highlight how a market's psychographics shift, point to alternate uses of emerging technology, and detect other patterns from real-time data. As an example, AI is used for sentiment analysis to gain "useful insight[s] from unstructured and unorganized textual contents from several social platforms and online sources such as chats happening across social platforms such as Twitter, WhatsApp, and Facebook, online blogs and comments" (Yi and Liu, 2020: 621). Entrepreneurs can use these insights (as outcomes of AI) to learn about customers' opinions about a product or service, innovative ideas for addressing current problems, and people's reactions to current changes in the external environment.

Furthermore, AI can augment entrepreneurs' interactions with a community of inquiry rather than simply relying on traditional methods to collect data (which can be used to train AI). AI can also be used as a method. For example, artificial swarm intelligence is well suited to augment decision making under high uncertainty from which knowledge emerges (e.g., that faced by entrepreneurs); it is more effective than traditional modes of pooling collective intelligence. This form of AI is based on biological models of the collective intelligence of swarms—schools of fish, flocks of birds, and swarms of bees. The notion is that a group (or a swarm) engages in self-organized interactions, amplifying the group's intelligence and reducing its errors as group members collaborate and converge on a solution that optimizes the group outcome. This mimicking of a swarm via AI makes the most of the distributed attention, self-organized decision making, and collective intelligence of a group of humans. For example, swarm intelligence based on flocks of birds has been used to fly drones as a coherent and coordinated group, and swarm intelligence based on schools of fish has been used to synchronize robots' movements (Rubenstein et al., 2014). Similarly, Metcalf et al. (2019: 87) offer the following explanation for how the swarm intelligence of bees is mimicked in AI for humans:

Hundreds of individual scout bees search a large area for potential sites, and then deliberate as a closed-loop system to select a location, with each participant influencing the collective decision of the group through body vibrations that encode the direction and distance to possible colony sites along with the strength of support for that site, as well as stop signals that inhibit other dancers. This process of real-time negotiation over competing signals continues until a quorum emerges that favors a particular site (Felin and Powell, 2016). Optimal decisions are reached by the swarm over 80% of the time (Seeley and Buhrman, 2001). Recent developments in interface design, networking, and AI processing make it possible to mimic the decision process used by honey bees and to facilitate human swarming. Such software permits humans to explore options collectively in a decision space through an interface, while the AI engine processes in real-time the input or behavioral data that humans provide. This produces an intelligent system composed of networked participants, where each participant contributes unique tacit and explicit knowledge in parallel with other participants, as the group makes a decision that is optimized for the collective.

Entrepreneurs can use artificial swarm intelligence to engage a community of inquiry about a potential opportunity; tap into the knowledge of the different members (potential users, suppliers, investors, employees, and so on); apply their unique tacit knowledge about aspects of the potential opportunity in parallel; and observe, cause, and interact with refinements of the potential opportunity by the collective in real time (see Majchrzak et al., 2021). For example, potential users of a product can provide ongoing information about how they want the product to work, how it feels, how it looks, how they identify with it, etc. The engineers of the product can provide ongoing information about what is technologically feasible, what shortcuts are possible, how the product could be made lighter/stronger/longer, etc. The production people can provide information about production costs, the lead time required, the level of customization provided at a high production volume, etc. The suppliers can provide information about the availability of different materials, the attributes of different sources of the raw materials, the costs of transportation, etc. The retail outlets can provide information about promotion, pricing, and potential distribution. This artificial swarm intelligence keeps entrepreneurs (and the rest of their communities of inquiry) in the loop of AI systems to facilitate the interactive process of developing and refining potential

opportunities.

We believe that artificial swarm intelligence will become an essential tool for augmenting a range of entrepreneurial tasks and is, therefore, an essential topic for entrepreneurship researchers. Moreover, as a research method, artificial swarm intelligence could provide researchers insights into the dynamic decision making of entrepreneurial teams and communities of inquiry as they co-construct potential opportunities or perform other social-based entrepreneurial tasks.

We recognize that many different factors shape opportunity attractiveness, and AI is likely to inform some factors of opportunity attractiveness more than others. As detailed above, there are numerous examples of how AI can notice changes in the external environment that signal a potentially attractive opportunity. However, entrepreneurs are unlikely to be entirely replaced by AI. For example, entrepreneurs are needed to form strong relationships with community members for their continued support and to “manage” the emotional aspects of these relationships. Although not replacing entrepreneurs, AI will become increasingly important as entrepreneurs rely on it to augment their decision making, assessments, and actions.

We similarly hope that entrepreneurship scholars adapt artificial swarm intelligence for use as a research method, collect and analyze the resulting data, and address new entrepreneurship research questions that were previously inaccessible.

5.2. AI augmenting activity-based entrepreneurship

Traditionally, some entrepreneurship research has focused on one action that represents entrepreneurship. One action can be entering self-employment (Bates, 1995; Sørensen, 2007), starting a new venture (Busenitz and Lau, 1996; Shaver and Scott, 1992), or—more generally—exploiting a potential opportunity (Foss et al., 2013; McGrath, 1999). Of course, with these one-off (for a single study) events, many activities lead to, constitute, and are generated by such entrepreneurial action. A practice perspective of entrepreneurship explores these activities as a micro-foundation of entrepreneurial action. Shepherd (2015) asks what activities promote, reflect, and are generated by an entrepreneur's personal motivation and prior knowledge to generate the entrepreneur's belief that he or she has identified a potential opportunity for someone and whether that opportunity is personally desirable and feasible for the focal entrepreneur to exploit. AI may augment entrepreneurs in the activities underlying a micro-foundation of entrepreneurial action to exploit a potential opportunity.

In particular, AI can augment entrepreneurs in creating new ventures. Indeed, the act of new venture creation comprises a series of activities that advance, constitute, and arise from organizational emergence. Organizational emergence refers to progress in creating a new venture (Shepherd et al., 2021). Some of the activities indicating emergence include a nascent entrepreneur hiring employees for the first time, making the first sale, and securing external financing (Reynolds and Miller, 1992; Tornikoski and Newbert, 2007; Newbert and Tornikoski, 2012).

Although important research has been conducted on this topic, the sequencing of the activities underlying patterns of organizational emergence remains unclear. This lack of clarity likely arises from the heterogeneity among venture startups and the entrepreneurs who lead them. That is, the sequencing and bundles of activities differ for different types of potential opportunities (e.g., high tech to low tech [e.g., Cefis and Marsili, 2011], environments (e.g., dynamism, complexity, and munificence [Baum et al., 2001]), markets (e.g., business to customer or business to business [Parry, 2020], and entrepreneurs (e.g., personality, prior knowledge, motivation, activity-based background, etc.). Models have been used to describe sequences of activities for non-routine work, such as creativity (Lindberg et al., 2022), as well as sequences of activities in product-development processes. For example, Salvato (2009) list describe several such sequences, including the “recipe-book,” “in-house adapted,” “externally adapted,” “externally driven,” and “recombinant” sequences. As an example, the recipe-book sequence of activities for new product development includes the following order of activities: (1) sets the basic guidelines for product development, in line with the designer's intentions and company targets; (2) contacts the designer to define details of the preliminary drawings; (3) complete internal activities and related external activities involving suppliers to develop the product design in line with the designer's directions and the firm's targets; (4) complete internal/external activities to define the production details and handle administrative issues, such as the cost of molds or components; (5) check on performed activities in the form of a revision, a process-checkup meeting, or both; (6) carry out further product-development phases both internally and externally with suppliers; (7) decide on product packaging; and (8) perform final checks (Salvato, 2009: 392).

AI can enhance these models specifically for identifying patterns of (the sequencing of) activities to augment entrepreneurs' decisions related to facilitating organizational emergence. Indeed, there has already been some progress in AI augmenting sequencing decisions. For example, in data mining, an AI system has been built (using long short-term memory recurrent neural networks) and trained on organizations' previous projects, activity schedules, and outcomes at various stages of the mining process and is used as a project-control system that automatically generates work schedules to validate existing work schedules and revise future work schedules based on real-time information (Amer and Golparvar-Fard, 2019). This AI could also be useful in determining opportunity advancement or termination given an organization's continually changing portfolio of potential opportunities (for entrepreneurial decision making related to project advancement or termination, see Bakker and Shepherd, 2017). Therefore, these examples of AI indicate that AI can be developed to augment entrepreneurial decision making to effectively sequence activities for organizational emergence, the composition of an organization's portfolio of potential opportunities, and other critical entrepreneurial tasks.

6. Discussion

6.1. Human challenges in developing and managing AI

There are numerous challenges with developing “good” AI. First, AI can optimize outcomes for a large sample at the expense of

specific individuals. Second, input data can be biased, thus making AI biased, and a common belief that machines lack bias exacerbates this error. Finally, AI's many parameters, complex weighting schemes, and adaptive learning make it difficult for humans (including designers) to understand how AI makes decisions. With a lack of understanding of how AI makes decisions, humans have difficulty explaining the process and fixing problems when they occur (Kane et al., 2021). Indeed, this AI augmentation can lead to causal ambiguity such that humans and AI become deeply interwoven, making it difficult to disentangle the different input sources (Zaggl and Pottbäcker, 2021).

Thus, humans are the ultimate interpreters of recommendations generated by the tool. Therefore, they need to be kept as accountable arbitrators of which AI recommendations should be acted upon and which should be ignored. This role is essential because at this point, humans are better than AI at interpreting and understanding abstract concepts from data and outcomes (Pohl, 2008); considering highly unique and contextual conditions (Frize et al., 2005); determining the range of possibilities of decisions (Zaggl and Majchrzak, 2022); finding and understanding synergies and patterns across domains (also known as general AI, which is limited in current AI [Krohn, 2019; Berk, 2021]); and noticing, expressing, and responding to emotions, such as empathy and compassion (Huang et al., 2019). Of course, humans also have limitations in making decisions in these contexts (limited cognitive and attentional resources), especially in the entrepreneurial context (Shepherd et al., 2017).

6.2. AI augmenting entrepreneurs' hot cognition

Shepherd (2015) highlights how cognition and emotion influence each other and calls for more research on this hot (i.e., emotional) cognition. This call takes an interesting twist when the role of AI is considered. Although emotions can facilitate entrepreneurial decision making and action (e.g., affect as information [Foo et al., 2009; Hayton and Cholakova, 2012; Welpe et al., 2012]), they can also bias decision making (for a review, see Zhang and Cueto, 2017). Because machines cannot feel (i.e., they focus on thinking and analytical tasks [Huang et al., 2019] people believe AI's outcomes are unbiased. However, AI can produce biased outcomes when the training data reflects a bias—namely, bias is cooked into the algorithms (Dastin, 2018; Raisch and Krakowski, 2021). We believe entrepreneurship research, which has already explored biases and heuristics as well as ethics is well situated to explore the issues of discrimination, fairness, equality, and other ethics-related constructs in entrepreneurs' development and use of AI.

However, what are the consequences of taking emotions out of the entrepreneurial decision-making process? Could an AI system for augmenting entrepreneurs' decision making have the unintended consequences of enabling fewer people to start ventures, terminating ventures earlier, or introducing substantial risk-reduction strategies? Do optimism and confidence and even overoptimism and overconfidence benefit entrepreneurs, ventures, and society? Hayward et al. (2010) argue that overconfidence can be beneficial because it can help generate resilience to failure (i.e., increase the likelihood that an entrepreneur will try again). What are the consequences of entrepreneurs losing the benefits generated from overconfidence through the use of AI? Alternatively, to what extent does AI reinforce an entrepreneur's decision making (based on the training introduced by the entrepreneur), which likely increases the entrepreneur's overconfidence? Much research is needed on how AI can augment the feelings associated with entrepreneurial decision making.

6.3. AI augmenting entrepreneurs' compassion

Compassion is “an interpersonal process involving the noticing, feeling, sensemaking, and acting that alleviates another person's suffering” (Dutton et al., 2014: 277). Compassion can lead individuals to create new ventures to alleviate others' suffering, such as the suffering of victims of natural (Shepherd and Williams, 2014; Williams and Shepherd, 2016, 2021) and man-made disasters (Mittermaier et al., 2021). However, there are numerous challenges in organizing willing and compassionate individuals into an effective entrepreneurial response (Majchrzak et al., 2007; Stallings and Quarantelli, 1985; Strandh and Eklund, 2018). How can individuals organize other compassionate individuals to alleviate victims' suffering effectively? How can they make do with the tacit resources that were not destroyed, coordinate with people when traditional communication channels are disabled, and provide help when many roads are obstructed? While entrepreneurs have found ways to organize compassion for many disasters, this compassion venturing can be improved.

While all disasters have some distinguishing features, there are likely basic patterns (of challenges and responses) across disasters such that AI can augment entrepreneurs' compassion venturing. For example, we described artificial swarm intelligence above and how it can augment an entrepreneur's co-creation of potential opportunities with a community of inquiry. Artificial swarm intelligence could be helpful in compassion venturing by engaging different community members (including entrepreneurs from various ventures) to optimize the response to a community devastated by an adverse event. For example, artificial swarm intelligence software can help explore resource-collection/-allocation decisions by providing an interface for capturing the collective intelligence of the different community members. The swarm's critical members are the different sets of locals—the local victims and the local responders (often, these local responders are also victims of the adverse event). AI can perform an important role in matching the many problems arising from an adverse event to the locals with the skills to address them to help the community heal (see Majchrzak and Shepherd, 2021).³

³ However, this importance of local members (victims and responders) could highlight a flaw in AI as an effective, compassionate tool. What happens to local members' input to artificial swarm intelligence when the adverse event that caused the suffering also cuts connectivity. Perhaps the role of governments and established organizations is to provide some form of rapid response for connectivity to enable AI to augment compassionate individuals' attempts to act entrepreneurially to help others.

6.4. The dark side of AI

Indeed, Kane et al. (2021) explain how machine learning (specifically and AI more generally) can lead to a dystopian world, which they call Infomania, where human users lose (some) control over their lives. Similarly, Kellogg et al. (2020) describe how AI provides employers the opportunity to control employees by directing (i.e., restricting and recommending the behaviors of), evaluating (i.e., recording and rating), and disciplining (i.e., replacing and rewarding) their workers. Kellogg et al. (2020) also indicate how workers (individually and collectively) resist AI control mechanisms, which they call algoactivism. Over and above the dark side of AI for employees, AI offers a potential dark-side mechanism for destructive entrepreneurial action.

Specifically, when entrepreneurial action turns out poorly, entrepreneurs (like other people) are likely to blame others and the external conditions (Hayward et al., 2006; consistent with the fundamental attribution error [Harvey et al., 1981]). Entrepreneurs can use blame to morally disengage from entrepreneurial action that causes harm, such as the pursuit of opportunities that cause damage to the natural environment (Shepherd et al., 2013) and harm to community members (Shepherd et al., 2022) or involve other unethical behaviors (e.g., bribery and corruption [Baron et al., 2018]). Specifically, when people engage in unethical behaviors, they often shift responsibility away from themselves to other agents (Bandura, 1990). For example, entrepreneurs who take actions that harm nature and local community members are often motivated to blame others for their actions as a means of disengaging their morals—that is, engaging in a harmful act without the self-sanctions that come with taking responsibility for that act (Baron et al., 2015; Shepherd et al., 2022, 2013).

Herein lies a potential problem with AI in entrepreneurship. As AI augments an entrepreneur's decision making and action (which feed back into the system), the machine and the human become highly intertwined (Kane et al., 2021; Zaggl and Majchrzak, 2022). With difficulty (if not impossibility) in separating the entrepreneur's influence from the machine's influence (and vice versa), there becomes a potential accountability gap (Mittelstadt et al., 2016)—it is difficult to assign responsibility for the decisions and actions. This accountability is particularly problematic when AI augments destructive forms of entrepreneurship. The use of AI provides the entrepreneur with readily available mechanisms for moral disengagement by shifting responsibility to the machine. Maybe such a claim of blame (on the machine) is valid. Perhaps it is not. But who knows? The entrepreneur likely does not understand how decisions are made given the human-machine intertwinement. Overall, we are concerned that AI may shield entrepreneurs from responsibility for the destructive societal outcomes of their entrepreneurial action (despite their personal benefits from doing so).

What are the consequences of harmful entrepreneurial action when an entrepreneur blames the machine or stakeholders cannot assign responsibility to the entrepreneur? We cannot punish the machine. How does this impact the pursuit of entrepreneurial opportunities that are destructive to society? On the one hand, AI can highlight the likelihood and extent of the destruction caused by exploiting a specific potential opportunity, therefore making the costs more salient and dissuading its exploitation. On the other hand, AI provides moral-disengagement mechanisms that enable entrepreneurs to (believe they can) avoid sanctions from destructive outcomes from exploiting opportunities that generate personal gains. We hope future research explores how entrepreneurs can maintain responsibility for their actions and overcome the possibility of avoiding sanctions (by themselves and others) by blaming the machine. Without this research and practical knowledge, we risk entrepreneurs more easily sidestepping their morals and responsibilities to engage in destructive entrepreneurial action. With increased knowledge from this future research, entrepreneurs can take responsibility for the AI they use, and others, including regulators (see our discussion of regulations and governance above), can ensure that entrepreneurs are held responsible for their decisions and action (regardless of their level of entwinement with the machine).

6.5. AI's implications for entrepreneurship scholars

- We have discussed the most common forms of AI. Others currently exist, and there are more to come. Entrepreneurship scholars working at the intersection of entrepreneurship and AI need to be specific about the type of AI they are investigating. What will become less acceptable over time is talking about AI as a specific construct. Instead, we need to consider it an umbrella term that covers a host of AI-based constructs (e.g., machine learning or deep learning) and AI-based products in the hands of entrepreneurs and other stakeholders.
- Although it is difficult to understand the algorithms underlying the various forms of AI that entrepreneurs develop or use, it is essential that we do not give up just because it is difficult. As scholars, we need to understand how the focal AI works in a specific entrepreneurship context. Otherwise, it becomes simply a black box. A black box is the antithesis of scholarship.
- Some of the age-old questions of entrepreneurship are applicable in this stream of research. Should we focus on a specific application of AI in a specific context to better understand the nuances of the processes and practices at the intersection of AI and entrepreneurship, or should we focus on a broader set of AI-entrepreneurship relationships for more generalized knowledge? The answer is yes. We need both. We need to be open to research anywhere along the continuum—from thick descriptions to generalized models tested on global databases.
- An advantage of AI is taking emotions out of entrepreneurial decision making. A disadvantage of AI is taking emotions out of entrepreneurial decision making. With their knowledge of the role of emotions in the entrepreneurial context, entrepreneurship scholars are well positioned to advance our understanding of AI in the feeling economy. That is, rather than focus solely on taking emotions out of the decision-making process to reduce bias and remove obstructions, entrepreneurship scholars need to incorporate emotions with AI to improve decision making.
- Scholars have developed social entrepreneurship as a substantial stream of research. These scholars can help advance knowledge on how entrepreneurs can use AI to create greater social value for specific stakeholder groups or society more generally. In this way, AI

can be advanced into the feeling economy to exploit opportunities to alleviate people's suffering and solve other social problems rather than to pursue opportunities that manipulate people's emotions for commercial gain at a social cost.

- As AI disrupts the economy and kills some careers, scholars can explore how entrepreneurship can serve as a mechanism that helps people transition out of their dying careers and into careers that help them satisfy their psychological needs (of competence, autonomy, and relatedness) and enable them to find meaning in their lives. We do not want to undermine AI's threat to workers in certain careers, but there are potential opportunities for a better life in these threats. Entrepreneurship can be the vehicle to a new career or the destination career that improves people's lives. Moreover, social entrepreneurship scholars can explore ventures created to help alleviate people's suffering in transitioning from the previous economy to the new one.
- Entrepreneurship scholars explore how entrepreneurs make decisions under uncertainty and are thus well positioned to explore how entrepreneurs use AI responsibly. They can also shine a light on those who use the causal ambiguity surrounding AI to engage in entrepreneurial action that causes destructive outcomes for stakeholder groups or society.
- Management scholars have explored many paths to effective governance. That expertise needs to be applied to how AI is governed and how it impacts entrepreneurial action.
- While AI takes over more of humans' tasks, research must explore how entrepreneurs are kept in the loop of the AI tools. Entrepreneurship research needs to explain how entrepreneurs and stakeholders remain engaged with an AI product rather than simply creating one and assuming the entrepreneurial process is complete.
- The entrepreneurial context presents ethical challenges, and there has been much important research on the ethics of entrepreneurship. This importance is magnified at the intersection of entrepreneurship and AI because the uncertainty remains high. There are considerable negative consequences arising from ethical lapses related to AI, and there is an opportunity for entrepreneurs to "dodge" responsibility for moral disengagement.

7. Conclusion

AI and its machine learning are exciting developments in the technology world. How AI transforms markets and societies depends in large part on entrepreneurs. Entrepreneurs can use AI to augment their decisions and actions in pursuing potential opportunities for productive gains (productive for entrepreneurs, stakeholders, the natural environment, and society). Indeed, we detailed different ways in which AI may augment entrepreneurship. However, these propositions require the hard work of theory building, theory elaboration, theory testing, and empirical theorizing. Moreover, there are many other opportunities for future research that we identified but did not list, and we are sure there are many more that we have not even contemplated.

We hope this paper encourages people to consider research at the nexus of AI and entrepreneurship. Just like the importance of collecting data across the full range of possibilities to train AI, we hope that future research includes the bright side of how AI augments the productive outcomes of entrepreneurial action and the dark side of how AI augments the destructive outcomes of entrepreneurial action. By progressing research at the AI-entrepreneurship nexus, scholars can help entrepreneurs use AI to augment their decisions and actions for productive results and avoid some of AI's dark-side aspects in the entrepreneurial context. Furthermore, future entrepreneurship research can contribute to the computer science literature by adding a little entrepreneurship to thinking about AI.

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